M N S R NITHIN

mnsrnithin9@gmail.com | +91 88867 44899 | github.com/nithin999989 | linkedin.com/in/nithin-m-205b3a212

Profile Summary

Software Engineer with 2+ years of experience building scalable, cloud-native backend systems and distributed microservices. Proficient in Core Java, Spring Boot, gRPC, and event-driven architectures on Google Cloud Platform. Proven track record of designing high-throughput services (**100K+ requests/day**), slashing latency by **60–70%** via caching strategies, and delivering org-wide platform libraries that improve reliability and developer productivity.

Technical Skills

Languages: Java (Core Java, OOP), Python, JavaScript, SQL
Backend & Frameworks: Spring Boot, Microservices, REST APIs, gRPC, Resilience4j, Pub/Sub (Kafka)
Databases & Caching: AlloyDB (PostgreSQL), BigQuery, Redis
Cloud & DevOps: GCP, Kubernetes, Docker, GitLab CI/CD, Git, Maven, OpenTelemetry, Workload Identity Federation
Practices: Agile/Scrum, SDLC, System Design, Code Reviews, Production Debugging, Distributed Tracing

Experience

CAMS **June 2024 – Present**
Software Development Engineer – I

Notification Service | *Spring Boot, Microservices, Redis, gRPC, Docker, Pub/Sub (Kafka)*

- Architected and developed Spring Boot microservices with multi-level Redis caching, slashing database calls by **60–70%** and accelerating API response times by **∼40%** under high-volume notification workloads.
- Designed and implemented REST and gRPC APIs to enable low-latency inter-service communication across distributed microservices handling high-throughput event streams.
- Engineered resilience patterns – rate limiting and circuit breakers (Resilience4j) – to prevent cascading failures during peak load and improve overall system fault tolerance.
- Contributed to all phases of the Agile lifecycle: sprint planning, daily stand-ups, code reviews, and GitLab CI/CD pipeline automation to ensure reliable deployments.

reCAPTCHA / Bot Detection Platform | *JavaScript, Spring Boot, Pub/Sub, BigQuery*

- Designed and built an end-to-end reCAPTCHA service with a frontend JavaScript module that captures behavioral signals for bot and fraud detection, serving **∼100K live requests/day**.
- Instrumented **20+ browser behavioral signals** (mouse dynamics, keystroke cadence, scroll patterns) client-side, streaming encrypted payloads to the backend for real-time bot scoring.
- Implemented a hybrid encryption architecture using **RSA-OAEP (SHA-256)** and **AES-256-GCM** to securely transmit sensitive behavioral data between frontend and backend services.

Platform Common Libraries | *Spring Boot, Redis, Pub/Sub, Maven, GCP*

- Built a Pub/Sub publisher pool that reuses connections per topic (instead of per-message), with exponential backoff retries and graceful shutdown – adopted org-wide, improving resource efficiency and message delivery reliability.
- Configured profile-based Redis TLS for GCP Memorystore: production enforces SSL with CA-certificate validation while local uses plain connections, eliminating accidental production access from development machines.
- Created a centralized exception handler (`@RestControllerAdvice`) covering **15+** Spring and custom exception types, delivering consistent, field-level error responses across all microservices and reducing debugging time.

Achievements & Impact

- Resolved **25+** critical production incidents, including: implementing refresh-token-based authentication to extend secure session lifetime from 30 minutes to several hours, fixing Redis TLS misconfigurations, mitigating Pub/Sub outages with graceful degradation, and eliminating duplicate message processing via idempotency keys.
- Partnered with the Infosec team to design and standardize **GCP Workload Identity Federation (WIF)** for keyless cloud authentication – now the organization-wide security standard. *Medium blog*
- Piloted **OpenTelemetry distributed tracing** (POC with SRE), propagating trace/span IDs across **10 microservices** for end-to-end observability; authored the integration guide that was adopted org-wide. *Medium blog*

Education

Seshadri Rao Gudlavalleru Engineering College **July 2020 – Apr 2024**
*Bachelor of Technology in Computer Science — **CGPA: 9.02/10***

Narayana Junior College **Jun 2018 – Jun 2020**
*Intermediate (MPC) — **93%***